

A Multiplexing Scheme for Engineered Neural Communication Systems

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ABSTRACT

Transmitting information in engineered neural communication systems is a promising solution for delay-sensitive applications in the internet of bio-nano things (IoBNTs). As widely used in wired and wireless communication systems, multiplexing could also improve channel transmission efficiency for neural communication systems. In this paper, we model a neural communication system for IoBNTs and propose a neural signal multiplexing scheme for this system. It is a frequency-division multiplexing (FDM) based multiplexing strategy. The multiplexing, demultiplexing scheme, as well as decoding method are presented. The performance of the proposed strategy is evaluated and analyzed in terms of bit error rate and mutual information rate. The work can help researchers better understanding the underlying mechanism of neural multiplexing and pave the way for the implementation of IoBNT applications.

CCS CONCEPTS

• Information systems → Information systems applications; • Computing methodologies → Machine learning; • Mathematics of computing → Coding theory.

KEYWORDS

Molecular communication, neural communication, multiplexing, neural coding.

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1 INTRODUCTION

Recently, neural communication is proposed as a communication interface between the inside and outside human body, to connect the Internet of Bio-Nano Things (IoBNTs) in the human body and

external devices [15, 16]. It is envisaged that digital signals are transmitted via the human nervous system. The idea of transmitting information through the human nervous system is not groundless. In the field of molecular communication (MC), some researchers have proposed that nanomachines can be integrated into neuronal network systems to treat neurological diseases, such as Alzheimer. To this purpose, researchers proposed some signal transmission models for nanomachine-to-neuron communication as a preliminary step for future medical applications [11]. In [17], the authors introduced various types of MC channels and networks inside the human body including neuro-spike communication channels. In [18], the authors analyzed the correlated neuronal activity evoked by a theoretical nanoscale neuronal stimulator named synaptic nanomachine (SnM). In [1], the authors even finished the first controlled information transfer through an in vivo nervous system by modulating digital data from macro-scale devices onto the nervous system of earthworms. Researchers believe that the nervous system, which is a wired network throughout the body, could be used to reliably control various devices connected to the body, such as body area networks and brain-computer interface devices.

As an efficient and fast communication system, the human nervous system can receive sensory input, quickly respond and output motion control signals. The nervous system has been studied for decades, mainly in the neuroscience field. For example, on the problem of how perceptual information is encoded in the nervous system, neuroscientists have proposed different kinds of neural coding hypotheses such as rate coding, temporal coding and so on [3, 9].

Multiplexing refers to the technology to realize the simultaneous transmission of multiple signals on a dedicated physical channel. It is widely used in wired and wireless communication systems to improve channel transmission efficiency. It is confirmed by experimental data and analysis that multiplexing also exists in neural communication systems [2, 10, 20, 25]. In [20], the authors gave reviews about multiplexed sensory neural codes based on multiple temporal scales. Although there are still several challenges, evidence is accumulating about the existence and advantages of neural codes using multiplexed temporal scales. In [14], the authors proposed a multiplexing strategy named “synchrony-division multiplexing”. That strategy shows cortical neurons may encode the intensity of low-contrast features into the rate of asynchronous spikes, whereas the high-contrast features are encoded by the timing of synchronous spikes. In [6], the authors explained at the neuron level how the brain processes multiple input stimuli simultaneously. They found that neurons exhibited activity fluctuations consistent with switching between individual responses at different time scales when faced with different stimuli input at the same time.

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Unlike multiplexing based on different time scales in the above research works, the authors in [2] proposed an oscillatory multiplexing strategy of population codes which expresses multiple stimulus attributes by oscillation modulation of the spatial distribution of firing rate in the neuron ensemble. Reference [19] proposed a neural signal multiplexing technology based on compressed sensing from the perspective of signal processing. In that strategy, compressed sensing technology is used to demultiplex the composite signal from a large number of presynaptic neurons.

The above-mentioned research focuses on studying the phenomenological neural signal multiplexing to elaborate possible multiplexing behavior of neurons. We note that multiplexing is also well studied in the communication engineering field, such as frequency-division multiplexing (FDM), time-division multiplexing (TDM) and so on. However, to the best of our knowledge, there is no relevant research on applying communication theory and tools to design or analyze neural signal multiplexing system. In this paper, we will, from the communication engineering perspective combined with the physiological neuron multiplexing mechanism, design an artificial neural communication system to realize multi-signal transmission on a single neuron channel. We propose a possible FDM strategy and carry out simulations to evaluate bit error rate and mutual information rate. Our work will not only help researchers better understanding the underlying mechanism of neural multiplexing system, but also pave the way for future implementation of in-vivo delay-sensitive IoBNT applications. The main contributions of this paper are:

- 1) A multiplexing strategy is proposed for an engineered neural communication system based on single neuron channel.
- 2) The performance of multiplexing and demultiplexing schemes is evaluated in terms of bit error rate (BER) and mutual information (MI) rate.

2 SYSTEM MODEL

In the proposed scenario, there are two pairs of transmitters and receivers, i.e., TX1 and RX1, and TX2 and RX2, sharing a common neuron channel (see Fig. 1). It is assumed that transmitters are neuron-like nanomachines that can induce spikes of the channel neuron through certain stimulus methods, such as electric current microstimulation.

The transmitter sends signal by evoking action potentials (APs) at the axon hillock (yellow cone-shaped bump where the cell body connects to the axon). The AP then propagate along the axon due to the consecutive open and close of the neighboring ion channels on the axon. Its waveform is shown at the top of Fig. 1. When the AP arrives at the end of the axon, it triggers the presynaptic membrane to release neurotransmitters into the synaptic cleft, then the postsynaptic membrane (receiver) can finally receive this signal.

It is worth noting that although the spike is the physical carrier of information in neural communication, information is not encoded by a single spike but a spike train [3]. Different patterns of the spike train could represent different information, which is named neural coding. In Section 3, we will give the detailed coding schemes of transmitters during the introduction of our proposed multiplexing scheme. Here, the general form of the signals in this system is presented. Both TX1 and TX2 use spike trains to encode information.

Assume a bit interval T is divided into N_s time slots. The unit slot duration is defined as Δt . Thus, T is equal to $N_s \Delta t$. Limited by the maximum firing rate of neurons [22], it is assumed that at most one AP can be generated in each slot. On-off keying (OOK) modulation is used for both transmitters. Ignoring the shape of the AP, consider using a Dirac delta function to represent a single AP. Thus, the signal during one bit interval transmitted from TX1 can be expressed as

$$s_1(t) = \begin{cases} \sum_{i=1}^{N_1} \delta(t - t_i), & x_1 = 1 \\ 0, & x_1 = 0 \end{cases}, \quad (1)$$

where t is the present time during a bit interval, t_i is the time when a spike is fired and N_1 is the number of spikes in this spike train. x_1 is the bit transmitted by TX1 in this bit interval. Similarly, the signal sent by TX2 during one bit interval is

$$s_2(t) = \begin{cases} \sum_{j=1}^{N_2} \delta(t - t_j), & x_2 = 1 \\ 0, & x_2 = 0 \end{cases}, \quad (2)$$

where t_j is the time when a spike is fired and N_2 is the number of spikes in this spike train. x_2 is the bit transmitted by TX2 in this bit interval.

For the noise model, neurons may generate spikes spontaneously, so the spontaneous spikes can be considered as the noise in the neural channel. Since the generation of spontaneous spikes are always modeled as Poisson random process in neuroscience. It can also be modeled as a Poisson process with noise density λ_0 [4, 24], which can be expressed as

$$n(t) = \text{Poiss}(\lambda_0). \quad (3)$$

It should be noted that the generation of APs follows an all-or-none principle, which states that the nerve fiber will always give a maximum response and producing spikes of the same amplitude when stimulated [13]. Even if the intensity of the stimulus increases, the height of the spike always remains the same. Therefore, if TX1, TX2, or noise happens to generate spikes during the same slot, there will be only one spike generated with the same amplitude as other spikes.

At the receiver side, there will be a filter for demultiplexing, followed by a decoding module. We assume that all transmitters and receivers in this system are synchronized. The signal flow of the entire communication system is shown in Fig. 2.

3 PROPOSED MULTIPLEXING & DEMULTIPLEXING STRATEGY AND DECODING METHOD

In this section, we will introduce multiplexing and demultiplexing strategy. The decoding method is also presented.

3.1 Multiplexing and Demultiplexing Strategy

The multiplexing strategies in telecommunication system include FDM, TDM, code division multiplexing (CDM) and so on. In neural communication system, similar strategies could also be utilized by different coding schemes of the spike train.

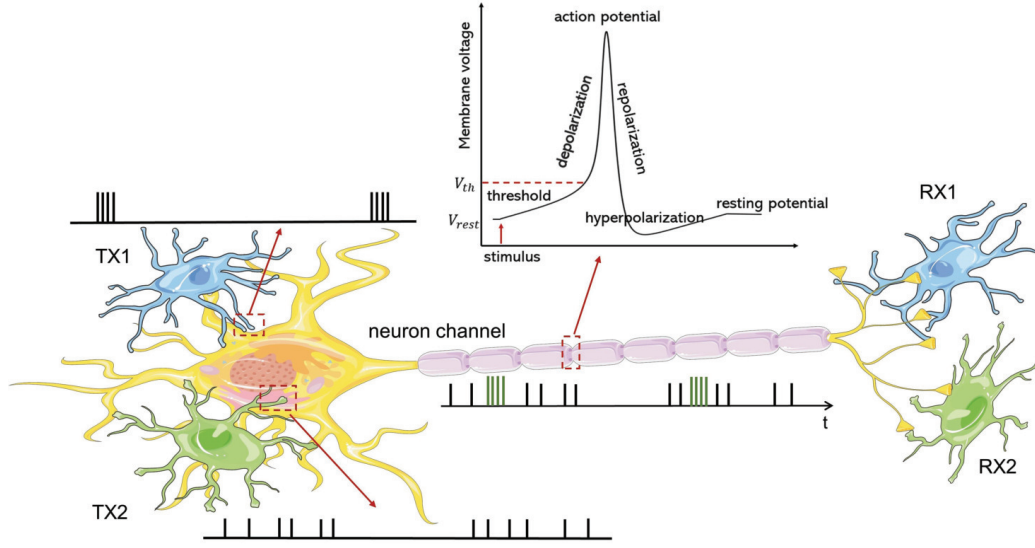


Figure 1: System model. TX1 and TX2 are two transmitters contact directly with the neuron. They use different coding strategies to encode information. The neuron in the middle is neuron channel which transmits multiplexed signals from transmitters. RX1 and RX2 are two receivers. The AP sequences evoked by TX1 and TX2 are both represented as δ pulse trains. The curve above the axon shows the waveform of the AP transmitted on axon.

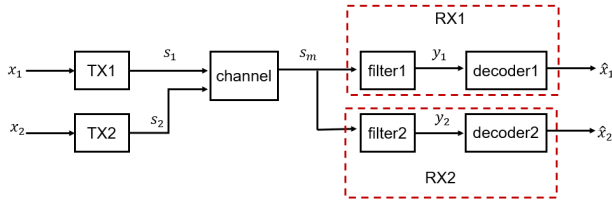


Figure 2: The signal flow of the proposed system. Signals from two transmitters are multiplexed in the same channel and are demultiplexed and decoded at the receiving end.

According to neuroscience literature, there are many hypothesized coding schemes, such as rate coding and temporal coding [3, 9]. Temporal coding refers to use accurate timing of APs or high-frequency fluctuations of firing rate to encode information. Rate coding, also called frequency coding, refers to the use of firing rate (the number of APs in unit time) to encode information.

We utilize two coding schemes for these two transmitters following the natural coding schemes of neurons. For TX1, it uses the spike burst to encode information, that is, a group of APs generated in rapid succession. More specifically, for TX1, bit “1” is represented by a spike burst which is constructed by N_1 consecutive spikes (N_1 in eq. (1) can be any physiologically possible value and is defined as a constant in our system). Assume that spike burst fires at the maximum firing rate of neurons. As mentioned in Section 2, Δt is exactly the absolute refractory period of neurons. Thus, interval between two consecutive spikes in spike burst is Δt . For (1), we have $t_2 - t_1 = t_3 - t_2 = \dots = t_{N_1} - t_{N_1-1} = \Delta t$.

For TX2, rate coding is applied which encodes information by the average number of spikes per unit time (also called firing rate) in

each bit interval. Usually in neuroscience the spike train of neurons is modeled as a Poisson random sequence [8]. So it is assumed that TX2 evokes a Poisson spike train in the time interval when bit “1” is transmitted. Thus, $s_2(t)$ in (2) can also be expressed as

$$s_2(t) = \begin{cases} \text{Poiss}(\lambda_1), & x_2 = 1 \\ 0, & x_2 = 0 \end{cases}, \quad (4)$$

$$\lambda_1 = \frac{E[N_2]}{\Delta t}, \quad (5)$$

where λ_1 is the firing rate, N_2 is the number of spikes during a fixed bit interval in (2), which is a random variable depending on firing rate. λ_1 represents the expected number of APs generated per unit time and it is also called the intensity of the Poisson process. According to the property of the Poisson process, λ_1 can be calculated by (5). Since the bit interval is divided into N_s slots, the probability of firing a spike in each slot during Δt is $\lambda_1 \cdot \Delta t$.

Fig. 3 shows the amplitude spectrum of two examples of $s_1(t)$ and $s_2(t)$ without DC component. It can be seen that the main frequency components of spike trains $s_1(t)$ and $s_2(t)$ are in different frequency bands, meaning that they can be separated from frequency domain. It is known that when neurons generate a spike, its membrane potential changes from negative potential to positive potential. What’s more, in real neural signals, spike burst contains high-frequency membrane potential oscillations, so it should be a high-frequency signal, whereas sparsely distributed spike trains contain lower-frequency membrane potential oscillations. Thus, the essence of multiplexing through the two encoding methods we described earlier is frequency division multiplexing.

We use

$$s_m(t) = s_1(t) \parallel s_2(t) \parallel n(t) \quad (6)$$

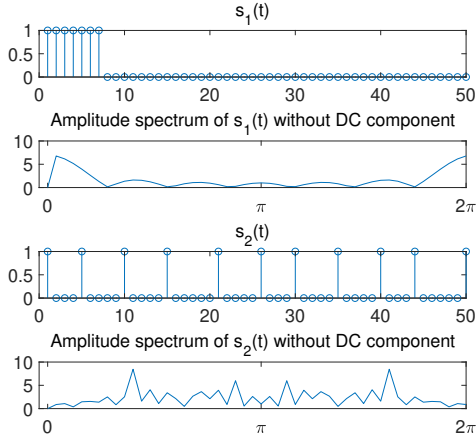


Figure 3: The spectrograms of two examples of $s_1(t)$ and $s_2(t)$ without DC component. $s_1(t)$ is a possible spike train for a bit “1” from TX1, and $s_2(t)$ is a possible sparse spike train for a bit “1” from TX2.

to express the superimposition of $s_1(t)$, $s_2(t)$ and $n(t)$. The notation of $\parallel$ indicates that the superimposition operation under the all-or-none principle as mentioned in Section II which can be realized by “or” logic operation. An example of multiplexing $s_1(t)$ and $s_2(t)$ is shown in Fig. 4. Sequence A [10011010101100010011] is encoded in signal $s_1(t)$. Sequence B [10011001011101011000] is encoded in signal $s_2(t)$. The multiplexed signal without considering noise spikes $s'_m(t)$ is shown at the bottom panel of Fig. 4. From Fig. 4, we can see clearly that when TX1 sends bit “1”, it can randomly generate a spike burst at any position of the bit interval. When TX2 sends a bit “1”, it generates a Poisson spike train during the whole bit interval. Some spikes in these two signals may superimposition with each other, but it is still very easy to distinguish these two signals since a spike burst contains very high frequency spikes in a short time compares to the entire bit interval.

$s_m(t)$ is received by both RX1 and RX2. The receiver performs demultiplexing and decoding tasks. Since the frequencies of the two spike trains $s_1(t)$ and $s_2(t)$ are different, demultiplexing can be easily achieved through filters. The filtering process can be implemented by organism [5, 7] or microcircuit. The demultiplexed signals of RX1 and RX2 denoted by $y_1(n)$ and $y_2(n)$ can be expressed as

$$\begin{aligned} y_1(n) &= F_1(s_m(n)) = \sum_{k=0}^N H_1(k)s_m(n-k), \\ y_2(n) &= F_2(s_m(n)) = \sum_{k=0}^N H_2(k)s_m(n-k), \end{aligned} \quad (7)$$

where $F_1(\cdot)$ and $F_2(\cdot)$ are filters for RX1 and RX2, respectively. $H_1(k)$ and $H_2(k)$ are the impulse responses of two filters that depend on specific design schemes. N is the impact length of input signal on filtered signal $y(n)$.

3.2 Decoding Method

After demultiplexing at the receiver, the signals are decoded to obtain the binary sequences. Neurophysiologists often investigate responses of a neuron by using the firing rate. Here we also use firing rate to reflect the property of spike train, thereby decoding the information. The firing rate, denoted as λ_z , is the frequency at which a neuron elicits APs. It is calculated by

$$\lambda_z = \frac{n_a}{\Delta t_z}, \quad (8)$$

where n_a is the spike number during the time period Δt_z . A spike burst means there are multiple spikes in a short period of time. In contrast, it is extremely difficult for a sparse spike train like $s_2(t)$ to have so many spikes in a short period of time. Thus, when the spike burst appears, the firing rate must increase rapidly. Since TX1 encodes information by the existence of the spike burst, RX1 needs to determine whether there is intensely change of firing rate in a bit interval. For RX2, it decodes the information by estimating the firing rate of the Poisson spikes in each bit interval and comparing its maximum with a predefined threshold. In this paper, kernel density estimation (KDE) is used to finish this task.

KDE is a classical method for estimating firing rate in neuroscience [23]. Different from the simple counting and interval division of the histogram, KDE weights and sums other data around each center sample according to the distance by convolving the data points with the kernel function. For spike firing rate estimation, KDE method is convolving spike train with the kernel function (such as Gaussian density function) to obtain multiple kernel functions. Next, they are linearly superimposed to get a smooth estimation of firing rate. This process can be formulated as [23]

$$KDE(z(t)) = \hat{\lambda}_z(t) = \int z(t-s)k_w(s)ds, \quad (9)$$

where $z(t)$ is the spike train whose firing rate needs to be estimated. $\hat{\lambda}_z(t)$ is the estimated firing rate and $k_w(s)$ is the kernel with the bandwidth w . For the KDE curve with multiple data points, the final curve shape is not closely related to the selected kernel function. There are several classic kernel functions for KDE algorithm, and the Gaussian curve is one of the most commonly used functions. In this paper, we also use Gaussian density function as the kernel which is

$$k_w(s) = \frac{1}{\sqrt{2\pi}w} e^{-\frac{s^2}{2w^2}}. \quad (10)$$

Then the decoding process of x_1 can be formulated as

$$\hat{x}_1(t) = \begin{cases} \text{bit "1"}, & KDE_1(y_1(t)) \geq th_a \\ \text{bit "0"}, & KDE_1(y_1(t)) < th_a \end{cases}, \quad (11)$$

where $KDE_1(\cdot)$ is kernel density estimation function for $s_1(t)$. th_a is the firing rate threshold of a burst. Thus, when $KDE_1(y_1(t))$ is larger than or equal to th_a , it proves that there is a spike burst in $y_1(t)$ and bit “1” is transmitted in $\{x_1\}$ in this bit interval. For RX2, the decoded signal is expressed as

$$\hat{x}_2(t) = \begin{cases} \text{bit "1"}, & KDE_2(y_2(t)) \geq th_b \\ \text{bit "0"}, & KDE_2(y_2(t)) < th_b \end{cases}, \quad (12)$$

where $KDE_2(\cdot)$ is kernel density estimation function for $s_2(t)$. th_b is the threshold of firing rate for no burst spike train. When firing

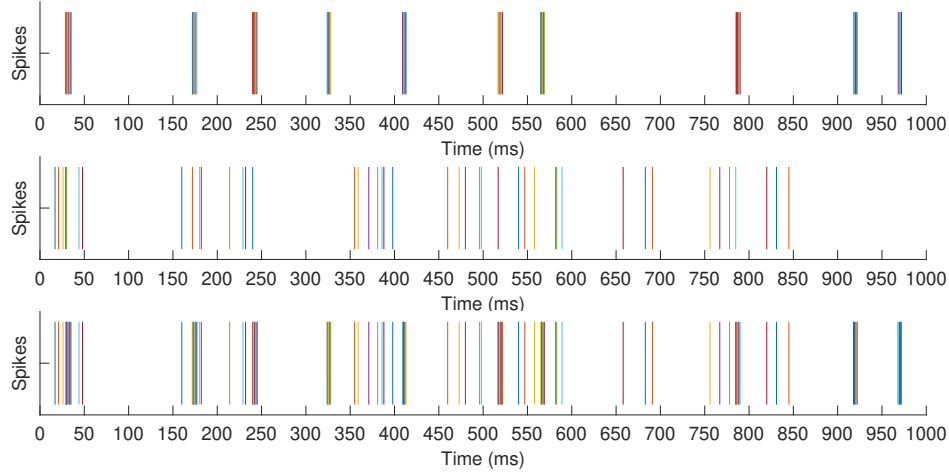


Figure 4: An example of multiplexing. The top panel represents the neuron spikes $s_1(t)$ encoding binary sequence A [10011010101100010011]. The middle panel is the neuron spikes $s_2(t)$ encoding sequence B [10011001011101011000]. The bottom panel shows the multiplexed signal transmitted in neural channel. Each bit interval lasts 50 ms. Both A and B sequence are 20 bits in length, thus the total time is 1000 ms in this example. λ_1 in (5) is set to 80 spikes/s, and spike number in a burst N_1 is set to 5 in this example.

rate $KDE_2(y_2(t))$ is larger than this threshold, it is determined that the symbol “1” is sent in $\{x_2\}$.

Two examples of decoding process of sequence A and B are presented in Fig. 5 and Fig. 6. As shown in Fig. 5, RX1 detects spike bursts in $y_1(t)$ by estimating the firing rate through KDE with a narrow Gaussian kernel. Since KDE with a narrow kernel can easily tract these high frequency spike bursts and there are only a few sparse spikes for each bit “1” in sequence B, sequence A can be directly decoded from the mixed signal through narrow-kernel KDE without filter in this example. For the decoding of sequence B, spike bursts must be filtered out. The process of filtering spike bursts is realized by the algorithm of finding the string since neural signal filter is out of our scope. The green curve shows the result of decoding sequence B by KDE with a broad kernel.

Here, we define the bandwidth of narrow kernel as 0.004 and the bandwidth of broad kernel is twice that of the former. The two thresholds should be selected according to the firing rate during encoding. In these two examples, we empirically choose 4.5 and 2.1, respectively. From Fig. 5, it can be seen that when the threshold and bandwidth are selected appropriately, the decision can be made correctly.

In fact, due to the randomness of the Poisson process and the spontaneous noise spikes, the number of spikes that encode the binary sequence is also random, which may cause decoding errors, especially for $s_2(t)$ when the spike distribution is relatively sparse. For example in Fig. 6, because the spikes in $y_2(t)$ are sparse in the eighth bit interval (0.35-0.4 s), the estimation density (green curve) can not exceed the threshold, which results in erroneous decoding.

4 SIMULATION RESULTS

In this section, the bit error rate (BER) and mutual information rate (MI rate) of the neural multiplexing system are simulated and evaluated by MATLAB.

The whole simulation process includes signal encoding (the binary sequence is converted into a neuron spike train), signal multiplexing, and signal decoding (the neuron spike sequence is converted to binary signal sequences). The data rate is set to 10 bits/s, that is, each bit interval is 0.1 s. There are 50 slots in one bit interval, i.e., $\Delta t = 2$ ms. λ_1 is set to 150 spikes/s, and λ_0 is set from 0 to 4 spikes/s. The above parameter selections are based on [21], the maximum frequency of APs that nerve fibers can transmit does not exceed 1000 Hz. 10^4 random generated bits are transmitted from TX1 and TX2. We repeat the simulation 10 times and average the results. In this simulation, we use the algorithm for finding strings instead of filters to extract or filter out high-frequency spike bursts to separate multiplexed signals since designing a real neural signal filter is out of our scope. Whereas in organisms, some neurons are inherently more sensitive to APs of a certain frequency, so filtering can occur naturally if the receiver is a neuron or a nanomachine with similar electrical characteristics [12]. Set thresholds th_a and th_b as 300 spikes/s and 90 spikes/s respectively. The spike burst length N_1 is defined as 8 spikes.

The simulation result of BER is carried out in Fig. 7. The probabilities of sending bit “1” and bit “0” are both 0.5. The bit error rate is simulated when multiplexing channel and when not multiplexing respectively. From the figure we can see that when only transmitting one sequence, $\{x_1\}$ with spike bursts, the BER is extremely low. When only using sparse Poisson spikes to encode information, the BER is slightly greater than using spike burst. If two transmitters send signals simultaneously, the BER of both signals increases

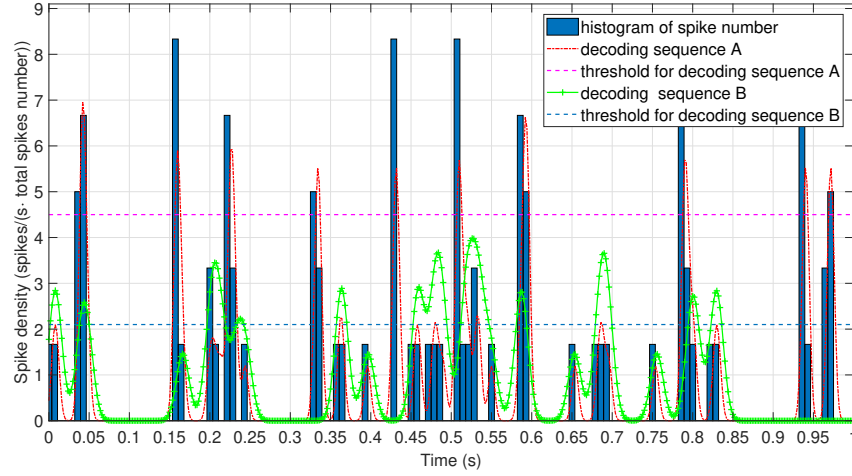


Figure 5: The decoding process of two signals. The bit interval T equal to 0.05 s, which is exactly the interval between two coordinate values on the horizontal axis. Red curve shows the KDE curve of sequence A [10011010101100010011] transmitted from TX1. When the red curve exceeds the purple horizontal line (threshold), it is considered that there is a spike burst in this bit interval, that is, bit “1” is sent. Green curve shows the KDE curve of sequence B [10011001011101011000] transmitted from TX2. When the green curve exceeds the blue horizontal line (threshold), it is considered that bit “1” is transmitted in current bit interval.

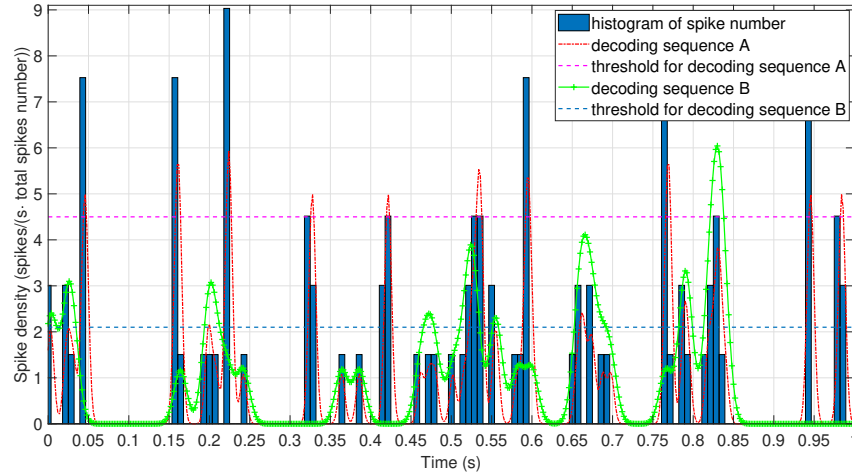


Figure 6: An example of decoding process of two signals and there is one bit error in $\{\hat{x}_2\}$ at the eighth bit interval between 0.35 s and 0.4 s. The green curve which represents the KDE curve of $y_2(t)$, does not exceed the threshold in the eighth bit interval. The bit interval T equal to 0.05 s, which is exactly the interval between two coordinate values on the horizontal axis. A sequence is [10011010101100010011] and B sequence is [10011001011101011000]. The $y_2(t)$ spikes are sparsely distributed in the eighth bit interval, and bit “1” is incorrectly decoded as bit “0”.

compared to when not multiplexing. In addition, the method of encoding information with sparse spikes is more susceptible to the influence of noise intensity, and its BER becomes larger as the noise intensity becomes larger.

The simulation result of MI rate is presented in Fig. 8. It can be seen that multiplexing slightly reduces the MI rate of each sequence. This is consistent with the conclusion that multiplexing increases the BER of the two signals in Fig. 7. Nevertheless, the total mutual information rate has increased significantly after channel

multiplexing as indicated by the top curve in the figure. In general, although it will introduce some errors, multiplexing still greatly improves the transmission efficiency of the channel.

5 CONCLUSION

In this paper, we propose a multiplexing neural communication system. The efficiency of the multiplexing strategy is evaluated by its BER and MI rate performance. The results show that this

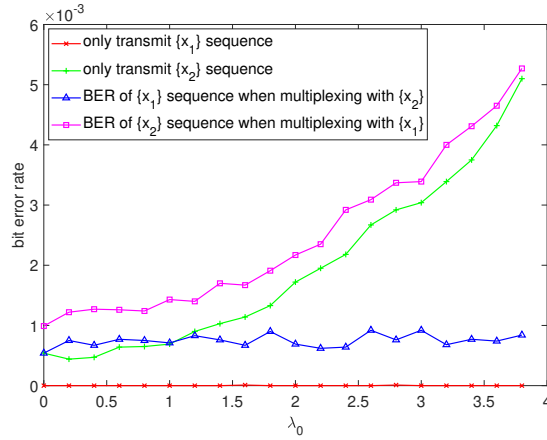


Figure 7: The BER of transmitting $\{x_1\}$ and $\{x_2\}$ sequence with and without multiplexing with respect to noise intensity λ_0 . $\{x_1\}$ sequence is the bits sequence transmitted by TX1, and $\{x_2\}$ sequence is the bits sequence transmitted by TX2. They are all automatically and randomly generated in the simulation program.

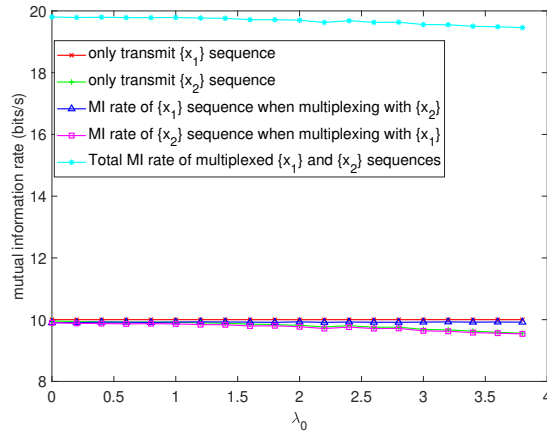


Figure 8: The MI rate of transmitting $\{x_1\}$ and $\{x_2\}$ sequences with and without multiplexing with respect to noise intensity λ_0 and the total MI rate in multiplexing communication system. $\{x_1\}$ sequence is the bits sequence transmitted by TX1, and $\{x_2\}$ sequence is the bits sequence transmitted by TX2.

system can work effectively with low BER. This work can help researchers better understanding the underlying mechanism of neural multiplexing and pave the way for the implementation of IoBNT applications. The aforementioned FDM strategy also has the potential to support more signals, as long as we can design multiple spike train patterns with different frequency distributions, which may provide the possibility of coexisting with original neurophysiological signals.

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